

Experiment No 04

Experiment Name

Write a program to match whether a given string is generated by a given pattern or not.

Objective

The objective of this experiment is to design and implement a program that checks whether a given input string matches a specified regular expression pattern. This experiment helps in understanding pattern matching, regular expressions, and their role in lexical analysis and compiler design.

Algorithm

1. Start the program.
2. Read the number of test cases.
3. For each test case:
 - (a) Read the regular expression pattern (R).
 - (b) Read the input string (S).
 - (c) Create a regular expression object using the given pattern.
 - (d) Match the input string against the regular expression.
 - (e) If the string fully matches the pattern, display "Match Found".
 - (f) Otherwise, display "No Match".
4. Stop the program.

Source Code

```
1  #include <iostream>
2  #include <regex>
3  using namespace std;
4
5  int main() {
6      int t;
7      string pattern, str;
8
9      // Read number of test cases
10     cout << "Enter number of test cases: ";
11     cin >> t;
12
13     while (t--) {
14         // Read regex pattern
15         cout << "Enter regular expression: ";
16         cin >> pattern;
17
18         // Read input string
19         cout << "Enter string to match: ";
20         cin >> str;
21
22         // Create regex object
23         regex r(pattern);
24
25         // Match string with regex
26         if (regex_match(str, r)) {
27             cout << "Match Found" << endl;
28         } else {
29             cout << "No Match" << endl;
30         }
31     }
32
33     return 0;
34 }
35
```

Figure 1: Source code for string pattern matching using regular expressions

Output Screenshot



```
1  SAKIB  Code
2  cd "c:\PU Projects\PUC Courses\8th Semester\CC Lab\Lab 04\Code\" ; if ($?) {
3  g++ final.cpp -o final } ; if ($?) { .\final }
4  Enter number of test cases: 1
5  Enter regular expression: a*bb
6  Enter string to match: aaabb
7  Match Found
```

Figure 2: Output showing whether the string matches the given pattern

Discussion

This experiment demonstrates the use of regular expressions to verify whether a given string is generated by a specified pattern. The program successfully checks complete string matches using built-in regex libraries. This approach simplifies lexical pattern recognition and provides insight into the use of regular expressions in compiler design and text processing applications.